

OSI LAYER

1. Draw a diagram of the OSI 7-layer model and label each layer with its correct name and a real-world example of a device or protocol that operates at that layer (e.g., Switch at Data Link, HTTP at Application).

7. Application Layer
6. Presentation Layer
5. Session Layer
4. Transport Layer
3. Network Layer
2. Datalink Layer
1. Physical Layer

7. Application Layer

Function

- It is responsible for providing an interface to user to interact with application.

Protocols

- HTTP (HYPER TEXT TRANSFER PROTOCOL)

- HTTPS (HYPER TEXT TANSFER PROTOCOL SECURE)
- FTP (FILE TRANSFER PROTOCOL)
- SMTP (SIMPLE MAIL TRANSFER PROTOCOL)
- DNS (DOMAIN NAME SYSTEM)

Example

Opening Instagram or Google in a web browser.

6. Presentation Layer

Function

- Responsible for defining standard format of data
- Formats, encrypts, decrypts, and compresses data.

Protocols

- SSL/TLS
- JPEG
- MP3

Example

HTTPS encrypts data before it is sent over the Internet.

5. Session Layer

Function

- Used to start maintain and terminate the session.
- Creates, manages, and ends communication sessions between devices.

Protocols

- NetBIOS
- RPC

Example

A WhatsApp voice or video call session.

4. Transport Layer (Layer 4)

Function

- Delivers data between application in network
- Breaks data into segments and reassembles them.

Protocols

- TCP
- UDP

Example

TCP ensures a file downloads without missing data.

3. Network Layer (Layer 3)

Function

- Finds the best path for data.
- Uses IP addresses to route data between networks.

Devices

- Router

Protocols

- IP
- ICMP

Example

A router sending data from your home network to the Internet.

2. Data Link Layer (Layer 2)

Function

- Transfers data between devices on the same network.
- Uses MAC addresses.

- Detects and corrects transmission errors.

Devices

- Switch
- Bridge

Protocols

- Ethernet
- PPP

Example

A switch sending data to the correct computer in a LAN.

1. Physical Layer (Layer 1)

Function

- Transmits raw bits (0s and 1s) through cables or wireless signals.
- Deals with hardware and physical connections.

Devices

- Hub
- Repeater
- Ethernet Cable
- Fiber Optic Cable

Protocols/Standards

- Ethernet
- USB

Example

Connecting a computer to a switch using an Ethernet cable.

2. Match the following networking scenarios to the correct OSI layer: a) Encrypting data before sending, b) Routing packets across networks, c) Converting data to electrical signals, d) Opening a web page in your browser, e) Detecting and correcting errors in frames.
Hint: Write your answers as 'Scenario: Layer Name (Layer Number)'.

- Encrypting data before sending → Presentation Layer (Layer 6)
- Routing packets across networks → Network Layer (Layer 3)
- Converting data to electrical signals → Physical Layer (Layer 1)
- Opening a web page in your browser → Application Layer (Layer 7)
- Detecting and correcting errors in frames → Data Link Layer (Layer 2)

3. Pick any app you use daily (like WhatsApp, Instagram, or Zomato) and describe how at least three different OSI layers are involved when you send or receive data in that app.

When I send a message on WhatsApp, different OSI layers work together:

1. Application Layer (Layer 7): WhatsApp lets me type and send the message.
2. Presentation Layer (Layer 6): The message is encrypted using SSL/TLS to keep it secure.
3. Transport Layer (Layer 4): TCP ensures the message is delivered correctly without errors.
4. Network Layer (Layer 3): The IP protocol routes the message through the Internet to the receiver.
5. Data Link Layer (Layer 2): The Wi-Fi router or switch sends the data within the local network.
6. Physical Layer (Layer 1): The data travels as electrical signals through cables or radio waves over Wi-Fi.

4. Given this scenario: You are unable to access a website, but you can successfully ping its IP address. Identify which OSI layer(s) might be causing the problem and explain your reasoning.

If I can ping the website's IP address but cannot open the website, the problem is most likely in the Application Layer (Layer 7) or the Presentation Layer (Layer 6).

Reason:

- The successful ping means the Network Layer (Layer 3) is working because the IP address is reachable.
- If the website does not open, there may be a problem with the HTTP/HTTPS service, the web browser, or the web server at the Application Layer.
- If there is an SSL/TLS certificate or encryption issue, the problem may be at the Presentation Layer.